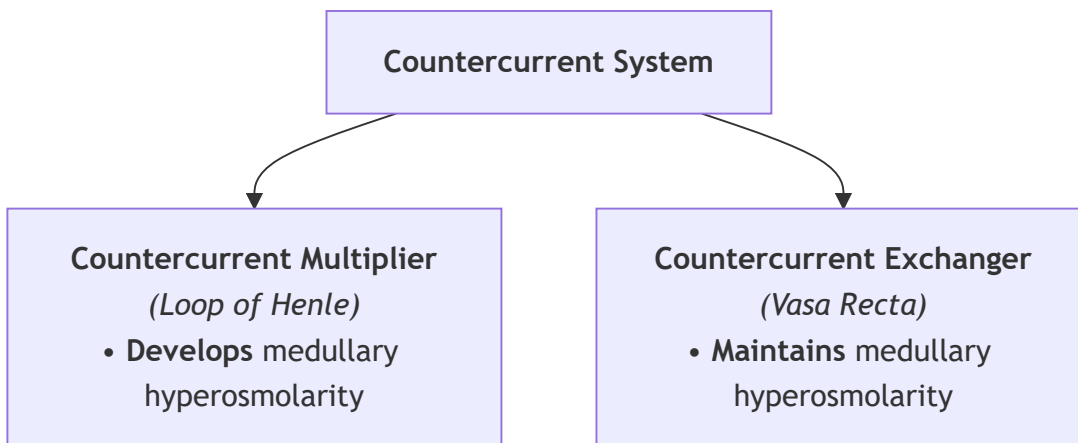


# COUNTERCURRENT MECHANISM

## Headings

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- **Definition** : A system of **U-shaped tubules (tubes)** in which fluid flows in **opposite directions** through the two parallel limbs of the loop.
- **Components of the Countercurrent System** :—
  - a. **Countercurrent Multiplier** : Loop of Henle
  - b. **Countercurrent Exchanger** : Vasa Recta

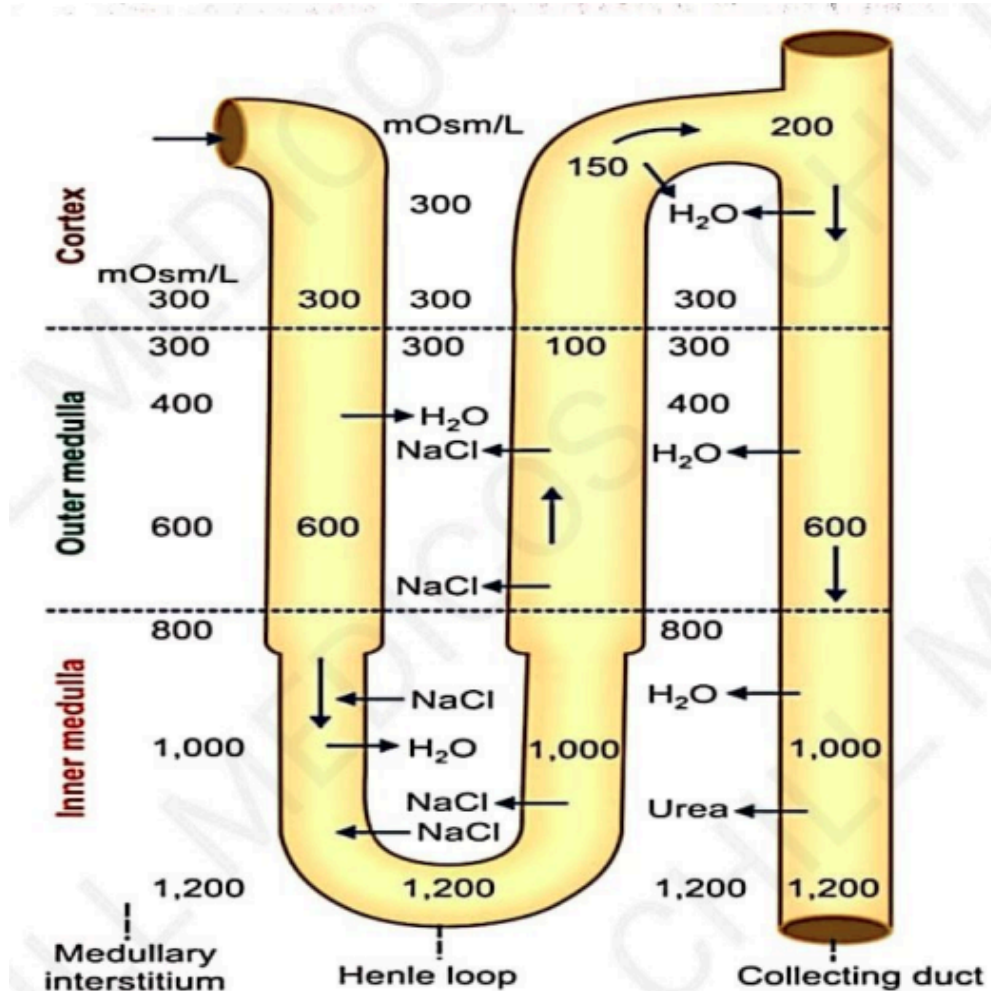


## 1. COUNTERCURRENT MULTIPLIER (LOOP OF HENLE)

- **Primary Role** : Responsible for the **generation/development of hyperosmolarity** in the medullary interstitial fluid and creating the vertical **medullary osmolar gradient** (300 to 1200 mOsm/L).

- **Role of Juxtamedullary Nephrons** : Play the major role because their loops of Henle are exceptionally **long and extend deep into the renal medulla**.

## Mechanism of Generating Medullary Hyperosmolarity



**FIGURE 53.1:** Countercurrent multiplier. Numerical indicate osmolarity (mOsm/L)

### **Ascending Limb of Henle**

- Active reabsorption of NaCl & other solutes (impermeable to H<sub>2</sub>O)



Solutes accumulate in  
**Medullary Interstitial Fluid**



↑ **Increases Medullary Osmolarity**  
(Creates high NaCl concentration gradient)



NaCl diffuses into  
**Descending Limb of Henle**  
& H<sub>2</sub>O is drawn out into interstitium



Filtrate becomes progressively concentrated at hairpin bend (1200 mOsm/L)



As concentrated filtrate enters ascending limb,  
**more NaCl is pumped out**



**Continuous Recirculation**

### Multiplies the Gradient

- **Solute Recirculation & Multiplication :**

NaCl and  $\text{Cl}^-$  ions are repeatedly recirculated between ascending and descending limbs



More solute is added to the tubular filtrate than is excreted



Multiplies the solute concentration in filtrate → **↑ Medullary Interstitial Osmolarity**

## Other Factors Responsible for Medullary Hyperosmolarity

### A. Reabsorption of Sodium from Collecting Duct

- In the medullary portion of the collecting duct, active  $\text{Na}^+$  **reabsorption** into the medullary interstitium further enhances interstitial hyperosmolarity.

### B. Recirculation of Urea

- The cortical collecting duct is relatively impermeable to urea.
- In the distal convoluted tubule (DCT) and cortical collecting duct (CT), water is reabsorbed under the influence of **Antidiuretic Hormone (ADH)** → **↑** increases the intraluminal concentration of urea.
- **Medullary Urea Diffusion :**

Concentrated urea reaches Inner Medullary Collecting Duct (IMCD)



Diffuses down its concentration gradient into the medullary interstitium



**Urea recirculation accounts for 50% of total hyperosmolarity in the inner medulla**

- **Urea Transporters :** Mediated by specific transporters ( $\text{UT-A}_1$  and  $\text{UT-A}_3$ ), which are activated and upregulated by ADH.

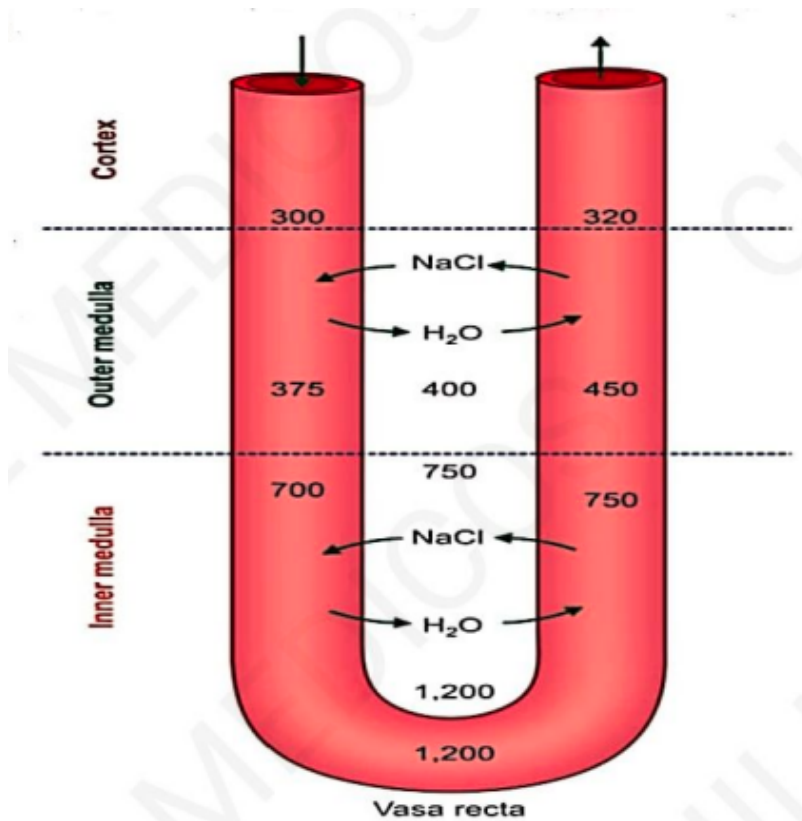
## 2. COUNTERCURRENT EXCHANGER (VASA RECTA)

- **Primary Role** : Responsible for the **maintenance of the medullary gradient** and hyperosmolarity created by the multiplier.
- **Anatomical Alignment** : Vasa recta acts as an exchanger due to its U-shaped hairpin loops arranged alongside the loop of Henle:
- **Descending limb of vasa recta** runs parallel to the **ascending limb of Henle**.
- **Ascending limb of vasa recta** runs parallel to the **descending limb of Henle**.

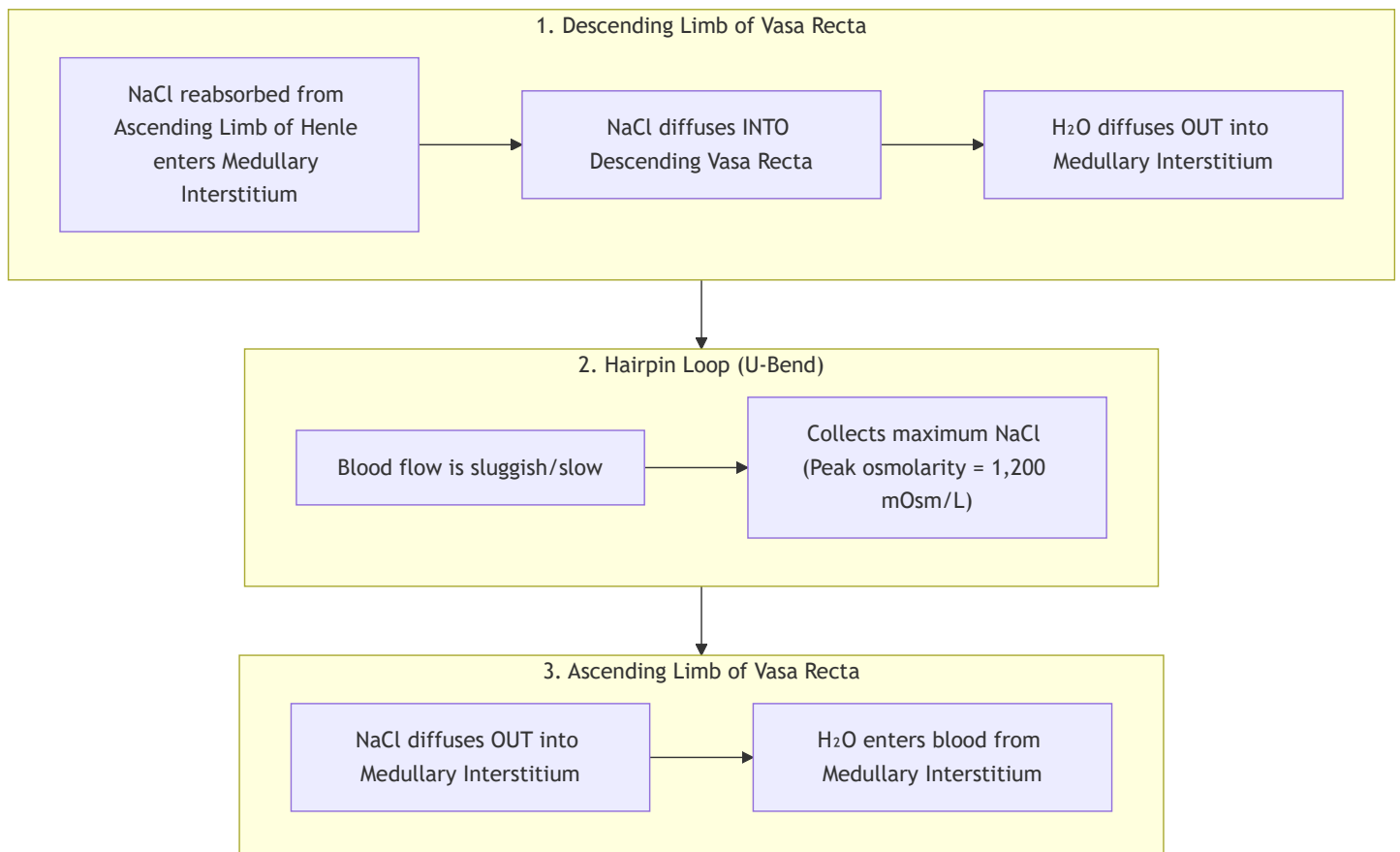
### OSMOLARITY GRADIENT ACROSS VASA RECTA

| Renal Zone         | Descending Limb (Arterial Side) | Medullary Interstitium | Ascending Limb (Venous Side) |
|--------------------|---------------------------------|------------------------|------------------------------|
| Cortex             | 300 mOsm/L                      | —                      | 320 mOsm/L                   |
| Outer Medulla      | 375 mOsm/L                      | 400 mOsm/L             | 450 mOsm/L                   |
| Inner Medulla      | 700 mOsm/L                      | 750 mOsm/L             | 750 mOsm/L                   |
| Hairpin Bend / Tip | 1,200 mOsm/L                    | 1,200 mOsm/L           | 1,200 mOsm/L                 |

# MECHANISM OF EXCHANGE



**FIGURE 53.2: Countercurrent exchanger.** Numerical indicate osmolarity (mOsm/L)



## Key Takeaway for Vasa Recta Function

- **Slow Blood Flow** : Sluggish medullary blood flow prevents the rapid "washout" of medullary hyperosmolar solutes.
- **Passive Exchange** :

**Vasa recta retains NaCl in the medullary interstitium and removes excess water**



**Maintains medullary interstitial hyperosmolarity**

- **Recycling of Urea** : Urea also diffuses into the descending vasa recta and exits via the ascending vasa recta, contributing to the continuous intra-renal recycling of urea.